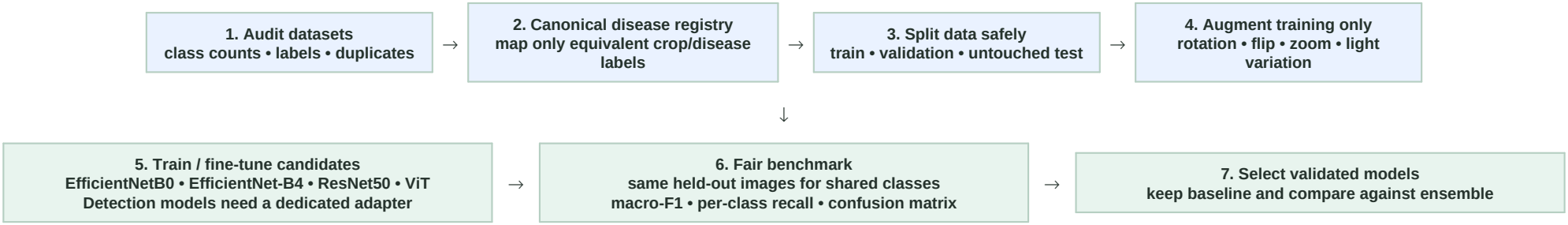
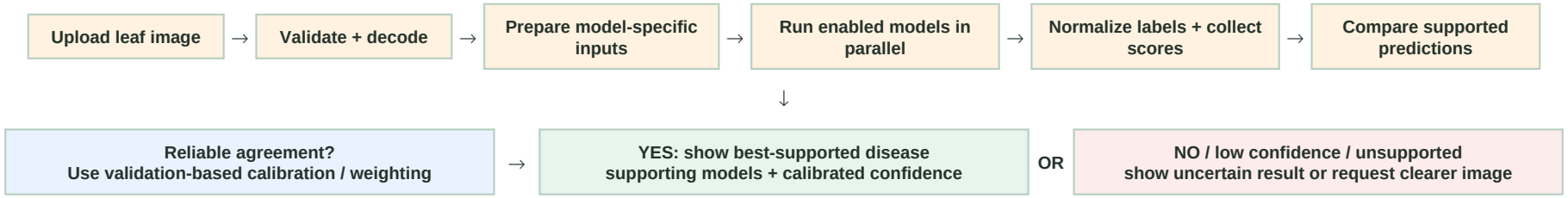
Plant Disease Prediction — Multi-Model Pipeline

Expanded datasets • model benchmarking • parallel inference • evidence-based prediction



PREDICTION PIPELINE



Evaluation rule: keep test data untouched until final evaluation. Results reported by different model cards are not directly comparable.

Model, Dataset & Database References

Candidate references for verification and benchmarking; not a claim that every model is compatible or more accurate.

Candidate models

Model / role	Repository link	Notes
Existing baseline — EfficientNetB0 / 38 classes	https://huggingface.co/Nefflymicn/PlantVillage-plant-disease-detection	Keep as baseline; verify local weights and exact class mapping.
EfficientNet-B4 classifier	https://huggingface.co/Khawajaa/plant-disease-detector	Verify labels, preprocessing, license and evaluation protocol.
ResNet50 classifier	https://huggingface.co/Sami-06/PlantdocAI-ResNet50	PlantDoc-oriented; map only genuinely equivalent labels.
RT-DETRv2 object detector	https://huggingface.co/Madras1/plantdoc-rtdetrv2-leaf-disease-detector	Detection outputs need a dedicated inference adapter.
Vision Transformer / tomato specialist	https://huggingface.co/oshriagronov/tomadoc-mythos	Specialized crop coverage; verify exact class list.

Datasets

Dataset	Link	Purpose
PlantVillage	https://github.com/spMohanty/PlantVillage-Dataset	Broad disease classification; mostly controlled backgrounds.
PlantDoc	https://github.com/pratikkayal/PlantDoc-Dataset	Real-world backgrounds and field-like images.
Plant Pathology 2020 (FGVC7)	https://www.kaggle.com/competitions/plant-pathology-2020-fgvc7	Apple leaf disease specialist dataset.

Database / label registry: inspect the existing project database configuration first; preserve its actual database name and connection settings. Store a canonical disease-label registry and dataset manifest without secrets.

Benchmark models on the same untouched test images for shared classes. Evaluate specialized models separately on their supported classes. Do not treat raw scores from different models as directly comparable probabilities.